MAHATMA GANDHI UNIVERSITY, KOTTAYAM
FIRST SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)
MG1CCRBCA101 - DISCRETE MATHEMATICS

Duration: 2 hours Maximum Marks: 70

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Answer any five questions. Each question carries 2 marks

1. Find propositions from the given statements. Write the truth values of the propositions. (a) Chennai is the capital of Tamil Nadu. (b) $x + 2 = 11$. [U] [3]
2. Define the disjunction of two propositions and write the truth table. [K] [3]
3. Write the set of positive integers less than 10 in (a) Roaster form (b) Set builder form [U] [1]
4. Let A represent the set of all students at a university, and let B represent the set of all courses offered at the university. What is the Cartesian product of $A \times B$? [U] [1]
5. Define transitive relation. Give an example of a transitive relation. [K] [6]
6. Let R_1 and R_2 be the relations from $A = \{1, 2, 3\}$ to $B = \{1, 2, 3, 4\}$ where $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ and $R_2 = \{(1, 1), (1, 2), (1, 3), (1, 4)\}$. Find $R_1 \cap R_2$ and $R_1 \cup R_2$. [A] [6]
7. Show that the matrix $\begin{bmatrix} 0 & -1 & 5 \\ 1 & 0 & 3 \\ -5 & -3 & 0 \end{bmatrix}$ is skew-symmetric. [U] [7]
8. Let $A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$, find (a) A+B and (b) A-B [U] [7]

Part B
Answer any five questions. Each question carries 6 marks

9. Express the following propositions using p and q and logical connectives. (a) It is below freezing and snowing. (b)) If it is below freezing, it is also snowing. (c) It is not below freezing and it is not snowing. [K] [3]
10. Find the bit strings for the following expressions:
    (a) $1\,1000 \wedge (0\,1011 \vee 1\,1011)$
    (b) $(0\,1010 \oplus 1\,1011) \oplus 0\,1000$ [K] [3]
11. (a) Define the complement of a set.
    (b) Let A be the set of positive integers greater than 10 (with universal set the set of all positive integers). Find the complement of A. [U] [2]
12. Prove the Absorption Laws: (a) $A \cup (A \cap B) = A$. (b) $A \cap (A \cup B) = A$. [A] [2]
13. Find the matrix representing the relations $S \circ R$ where $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ [An] [6]
14. The relation $R = \{(a, b)/\text{a and b have same parents}\}$ on the set of all people. Show that it is an equivalence relation. [A] [6]
15. For matrices $A = \begin{bmatrix} 6 & 1 \\ 3 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 3 \\ 1 & 2 \end{bmatrix}$, verify that $detAB = (detA)(detB)$ [U] [7]
16. Find the inverse of the matrix $A = \begin{bmatrix} 2 & 4 & 1 \\ 1 & 2 & 1 \\ 3 & 4 & 2 \end{bmatrix}$. [A] [8]

Part C

Answer any three questions. Each question carries 10 marks

17. (a) Prove that $(p \vee q) \wedge (\neg p \vee r) \rightarrow (q \vee r)$ is a tautology.
(b) Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent. [A] [4]

18. Make use of rules of inference, show that the following argument is valid:
"If today is Tuesday, I have a test in Mathematics or Economics."
"If my Economics Professor is sick, I will not have a test in Economics."
"Today is Tuesday and my Economics Professor is sick."
Therefore "I have a test in Mathematics". [An] [5]

19. (a) Define composite functions.
(b) Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g? What is the composition of g and f? [A] [6]

20. If A = set of integers and R is defined as aRb if and only if $a - b$ is divisible by 5.
(a) Show that R is an equivalence relation.
(b) Find all distinct equivalence classes. [A] [6]

21. (a) Solve the following system of equations using the matrix method:
$a + b + c = 3, a + 2b + 3c = 4, a + 4b + 9c = 6$.
(b) If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ show that $A^2 - 5A + 7I = 0$, where I is the identity matrix of order 2. [A] [8]

22. Solve the following system of equations using the Cramer's rule:
$2a + 3b = 4, 3a + 2b = -4$. [A] [8]